DSP in VLSI
Homework 3 - Interpolator

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1. (Step 1) Show the real-part and imaginary-part output waveform $\hat{x}_1[m+\mu]$ after interpolation using linear interpolator and second-order polynomial interpolator to interpolate the sampled waveform in the region of $10 \leq m \leq 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. In addition, draw the absolute error, defined by $|x_1[m+\mu] - \hat{x}_1[m+\mu]|$ in the region of $10 \leq m \leq 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. (30%)

   Figure 1 shows the interpolated waveform and the absolute error of the linear and second-order polynomial interpolators for $x_1[m+\mu]$.

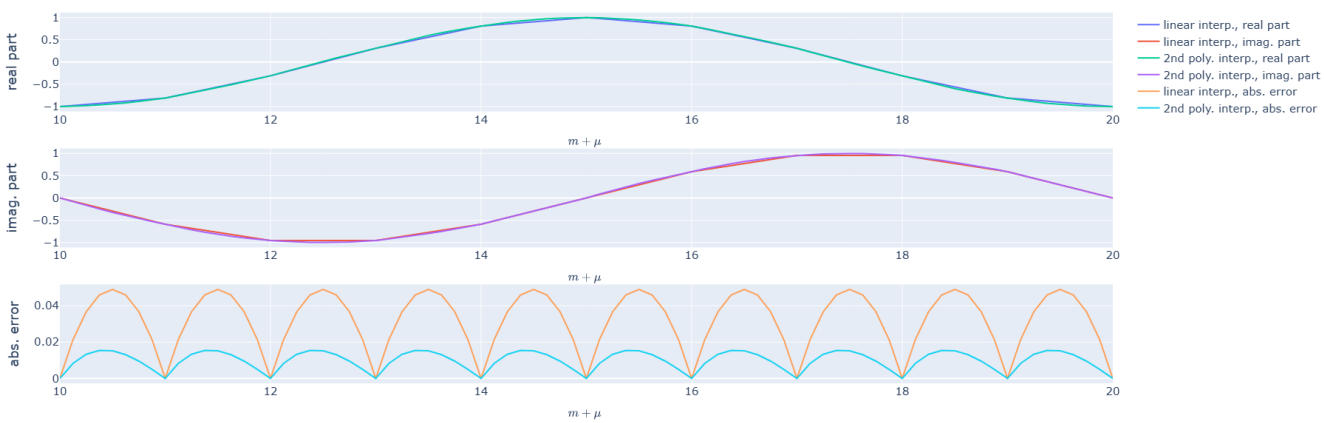

Figure 1: Interpolated waveform and absolute error of $\hat{x}_1[m+\mu]$

2. (Step 2) Show the real-part and imaginary-part output waveform of $\hat{x}_2[m+\mu]$ after interpolation using linear interpolator and piecewise parabolic interpolator to interpolate the sampled waveform in the region of $5 \leq m \leq 10$ with $\mu = 0, \frac{1}{9}, \frac{2}{9}, \ldots \frac{8}{9}$. In addition, draw the absolute error, defined by $|x_2[m+\mu] - \hat{x}_2[m+\mu]|$ in the region of $4 \leq m \leq 8$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. (30%)

   Figure 2 shows the interpolated waveform and the absolute error of the linear and piecewise parabolic interpolators for $x_2[m+\mu]$.

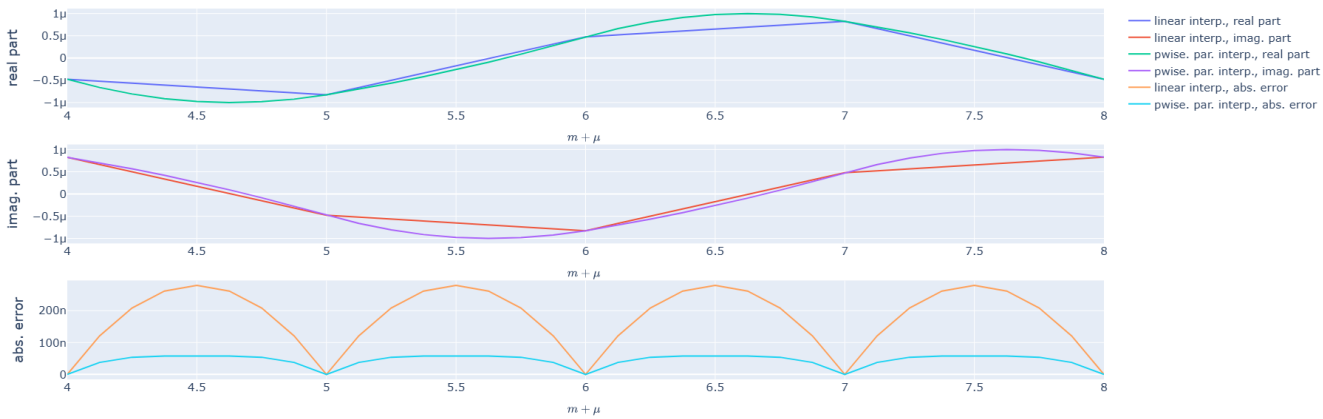

Figure 2: Interpolated waveform and absolute error of $\hat{x}_2[m+\mu]$

3. (Step 3) Please show the results calculated by your BF16 bit-true model for the following operands and operators. Express results both in decimal and binary representation. (S, E, F mean the sign bit, exponent field, and fraction field. All are given in binary.) (20%)

	Operand 1	**Operand 2**	**Operation**	**Result (Binary)**	**Result (Decimal)**
(a)	S: 1 E: 1000_0011 F: 0000_000	S: 0 E: 1000_1011 F: 0000_011	Addition	S: 0 E: 1000_1011 F: 0000_011	4192
(b)	S: 1 E: 0000_0001 F: 0000_011	S: 0 E: 0000_0001 F: 1111_010	Addition	S: 0 E: 0000_0000 F: 0000_000	0
(c)	S: 1 E: 0000_0010 F: 1100_000	S: 0 E: 0111_1100 F: 0000_110	Multiplication	S: 0 E: 0000_0000 F: 0000_000	0
(d)	S: 0 E: 0110_0011 F: 1011_110	S: 0 E: 1001_0011 F: 0101_000	Multiplication	S: 0 E: 0111_1000 F: 0010_001	0.00885009765625

Table 1: BF16 bit-true model computation results

4. (Step 4) Show the interpolation differences of the second-order interpolator using BF16 and double-precision arithmetic operations. Draw the figures indicating the difference of interpolated results caused by data format for outputs $\hat{x}_1[m+\mu]$ in the region of $10 \leq m \leq 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. (10%)

   Figure 3 shows the interpolated waveform of $\hat{x}_1[m+\mu]$ using double-precision and BF16 arithmetic, and the absolute difference between the two results caused by the finite precision of BF16.

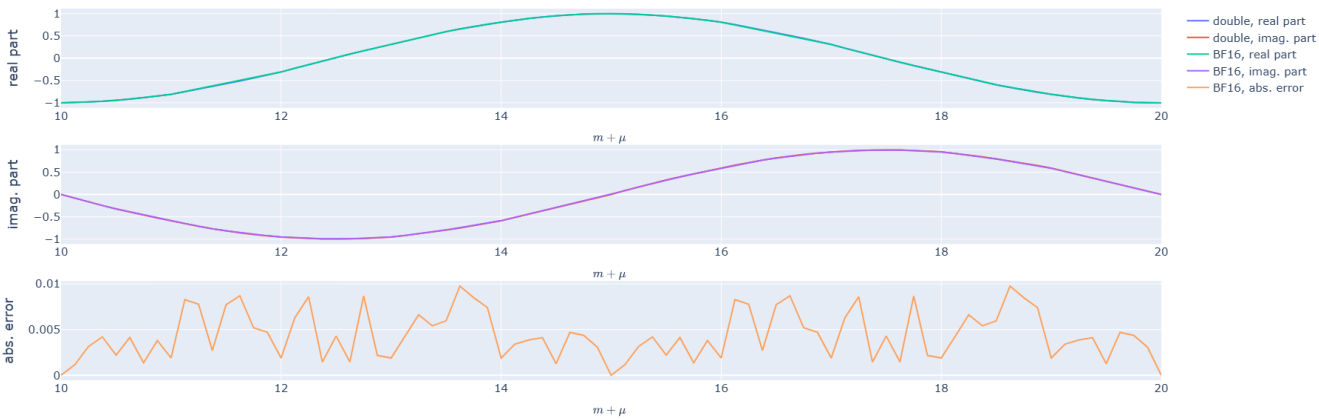

Figure 3: Interpolated waveform and absolute error of $\hat{x}_1[m+\mu]$ between double-precision and BF16

5. (Step 5) Draw the complete block diagram of your first version implementation according to the Farrow structure. (5%) Show your hardware sharing version (the precision of the datapath must be BF16 and cannot be changed). (10%) Draw the block diagram after your improvement. (5%) Indicate the critical path. (5%)

   Answer text here.

6. (Step 6) Write HDL to implement your second-order interpolator using BF16 arithmetic unit of Farrow structure after reduction. The input to be interpolated will change every 8 clock cycles and μ will change every clock cycle. Show the timing diagram of behavior simulation for the first 8 ($\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$) and the last 8 ($\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$) interpolated outputs of $\hat{x}_1[m+\mu]$ in the region of $10 \leq m \leq 20$. Show the real-part and imaginary-part error compared to double-precision results. (30%)

   Figures 4 and 5 show the timing diagrams for the first and last 8 interpolated outputs of the behavior simulation. Figure 6 shows the real-part and imaginary-part errors between the BF16 and double-precision results.

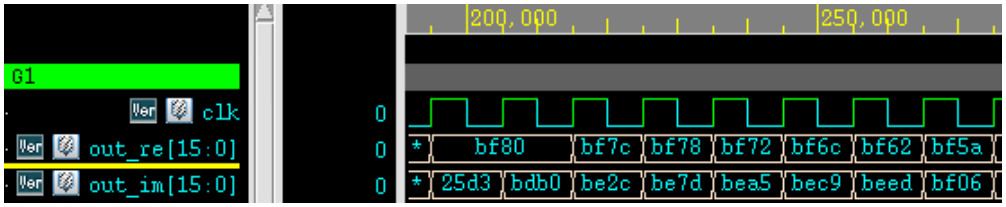

Figure 4: The first 8 output of interpolator

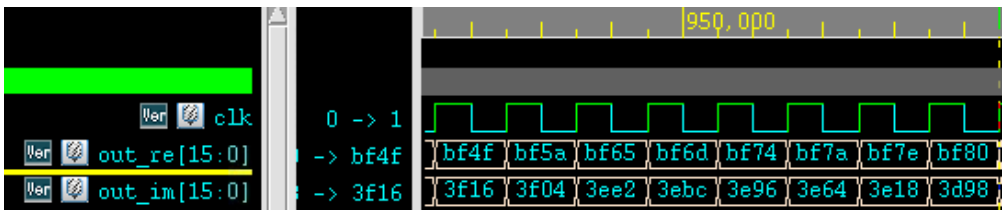

Figure 5: The last 8 output of interpolator

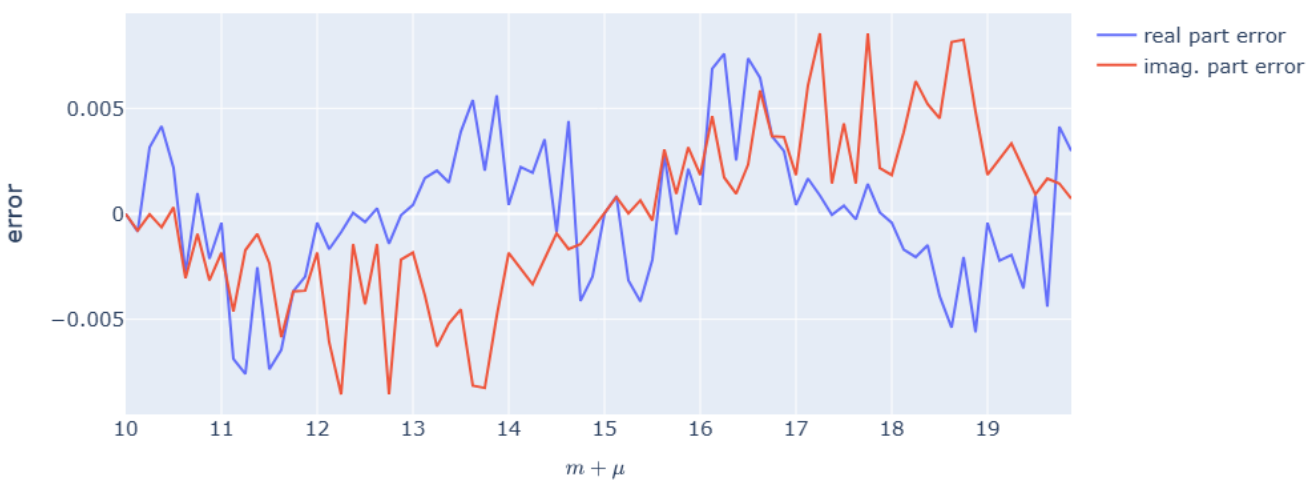

Figure 6: Real-part and imaginary-part error of $\hat{x}_1[m+\mu]$ between double-precision and BF16 in behavior simulation

7. (Step 6) Show your timing report to verify the critical path of your block diagram in Q5. (10%) List your timing constraint for post-synthesis simulation. (5%)

   The design is synthesized with a clock period of 20 ns (50 MHz). Input and output delays are set to 50% of the clock period. Figure 7 shows the timing report after synthesis.

```
Startpoint: in_valid (input port clocked by clk)
Endpoint: sr_im_reg[1][3]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
poly2_interpolator ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                     Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                     0.00       0.00
clock network delay (ideal)               0.50       0.50
input external delay                     10.00      10.50 f
in_valid (in)                             0.12      10.62 f
U2689/ZN (CKND1BWP20P90)                  0.15      10.77 r
U8473/ZN (ND2D1BWP20P90)                  0.03      10.79 f
U8474/ZN (OAI21D1BWP20P90)                0.01      10.80 r
sr_im_reg[1][3]/D (DFCNQD1BWP20P90)       0.00      10.80 r
data arrival time                                   10.80

clock clk (rise edge)                    20.00      20.00
clock network delay (ideal)               0.50      20.50
clock uncertainty                        -0.10      20.40
sr_im_reg[1][3]/CP (DFCNQD1BWP20P90)      0.00      20.40 r
library setup time                       -0.02      20.38
data required time                                  20.38
-----------------------------------------------------------
data required time                                  20.38
data arrival time                                  -10.80
-----------------------------------------------------------
slack (MET)                                          9.58
```

Figure 7: Timing report

8. (Step 6) Show the timing diagram of post-synthesis simulation for the first 8 ($\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$) and the last 8 ($\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$) interpolated outputs of $\hat{x}_1[m+\mu]$ in the region of $10 \leq m \leq 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. Show the real-part and imaginary-part error compared to double-precision results. (20%)

   Figures 8 and 9 show the timing diagrams for the first and last 8 interpolated outputs of the post-synthesis simulation. Figure 10 shows the real-part and imaginary-part errors between the BF16 and double-precision results, which are identical to the behavior simulation results.

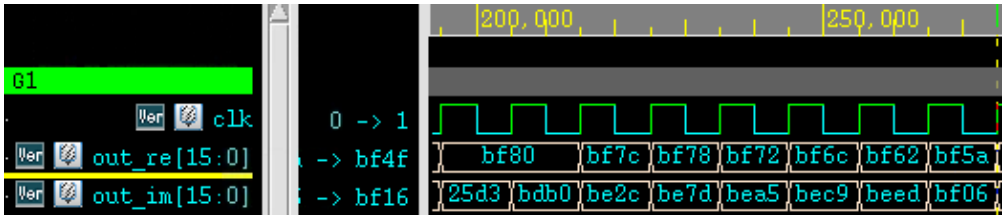

Figure 8: The first 8 output of interpolator (post-synthesis)

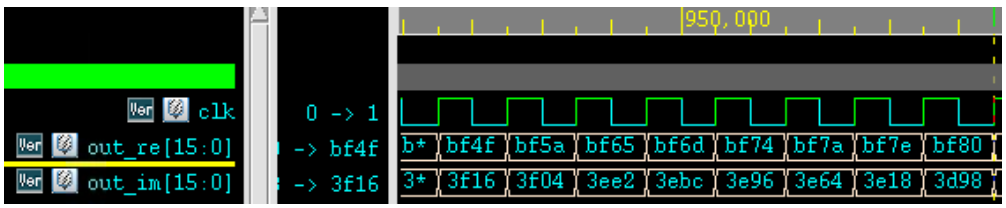

Figure 9: The last 8 output of interpolator (post-synthesis)

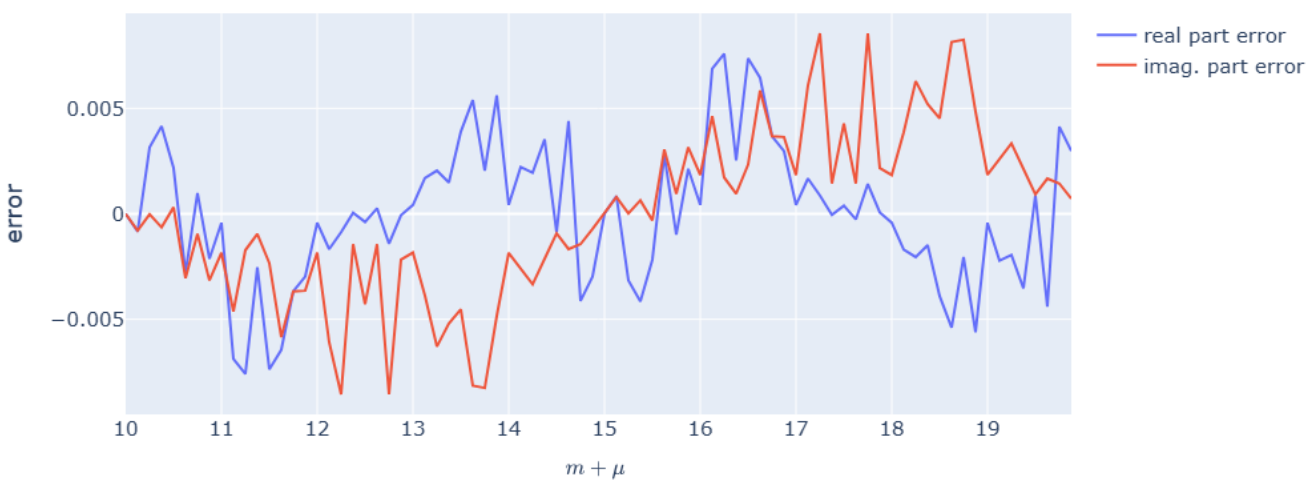

Figure 10: Real-part and imaginary-part error of $\hat{x}_1[m+\mu]$ between double-precision and BF16 in post-synthesis simulation

9. (Step 6) Construct your BF16 bit-true model as the architecture after your improvement. Depict the real-part and imaginary-part errors of interpolated outputs of $\hat{x}_1[m+\mu]$ between the Verilog outputs and the bit-true model in the region of $10 \leq m \leq 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. (20%)

   The Verilog outputs match the BF16 bit-true model exactly, so the errors are all zero as shown in Figure 11.

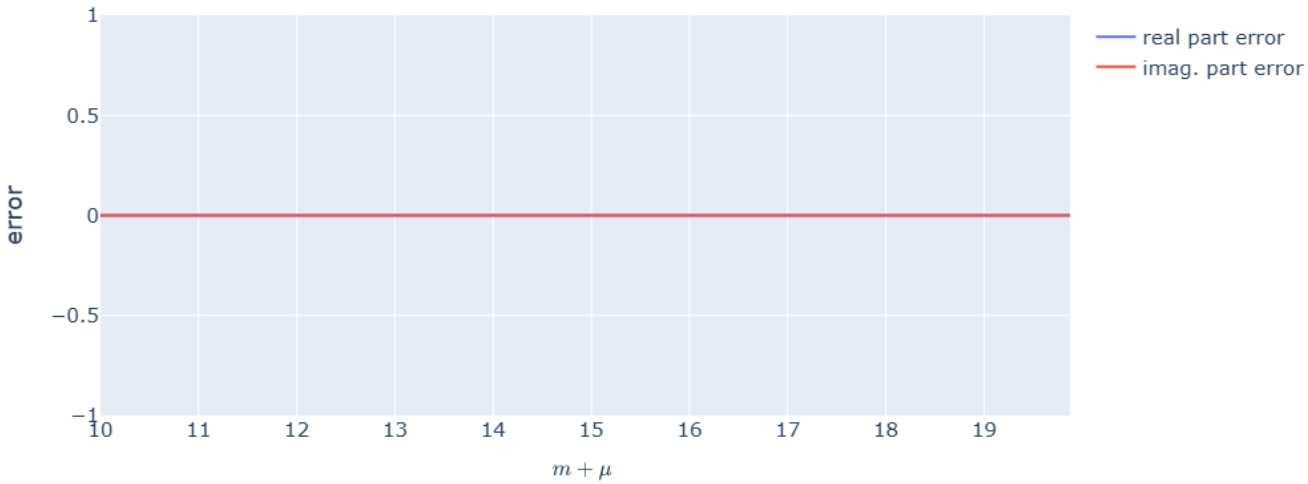

Figure 11: Real-part and imaginary-part error between Verilog outputs and BF16 bit-true model